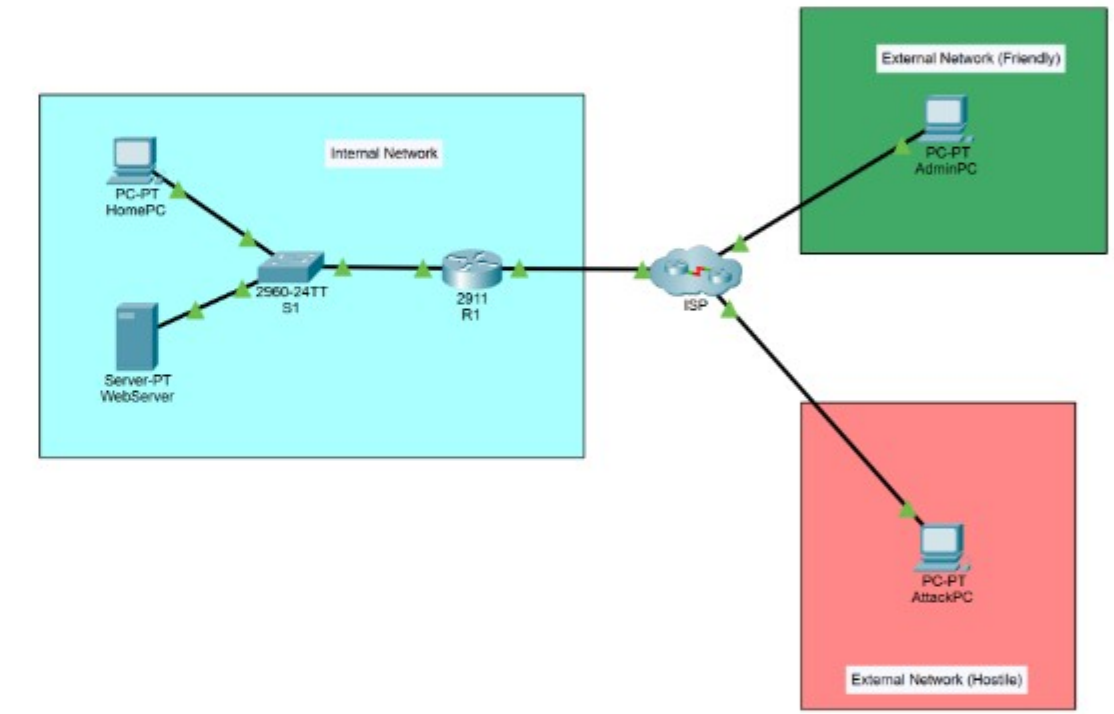
Open-Source Firewalls

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Introduction

- Firewalls: Smoothwall, OPNsense, and VyOS.
- Attack Combinations: Reconnaissance, Port Scanning, Vulnerability Scanning, Brute Force, Denial-of-Service.
- Desired Network Behaviour:
  - Web Server serve web page to WAN
  - SSH to Web Server from Admin PC
  - Outbound traffic from Web Server and Home PC



Network Topography

Smoothwall
- Default Behaviour & Added Rules

What It Is

- Simple, easy to use.
- Web-based.
- Key features:
  - Stateful Packet Inspection
  - Basic Filtering
  - Simple VPN

Default Behaviour

- No rules set.
- Result:
  - Blocks absolutely everything from the WAN_NET network.

Added Rules

- Allow Admin access to LAN_NET network.
- Allow ICMP traffic from WAN_NET to LAN_NET.
- Allow HTTP and HTTPS traffic from WAN_NET to Web Server.
- Allow SSH traffic from WAN_NET to Web Server.

Smoothwall - Tests

Reconnaissance

- Ping to Smoothwall's WAN_NET interface: successful
- Ping to Web Server: unsuccessful
- WAN_NET network scan (Nmap): successful

Port Scanning

- Port 80 forwarded from Web Server (Nmap): successful

Vulnerability Scanning

- Web Server scan (port 80, Nmap): successful
- Web Server scan (vulnerability, Nmap): successful

Brute Force Attack

- [With correct username & password] Web Server brute force (hydra): successful

Denial-of-Service

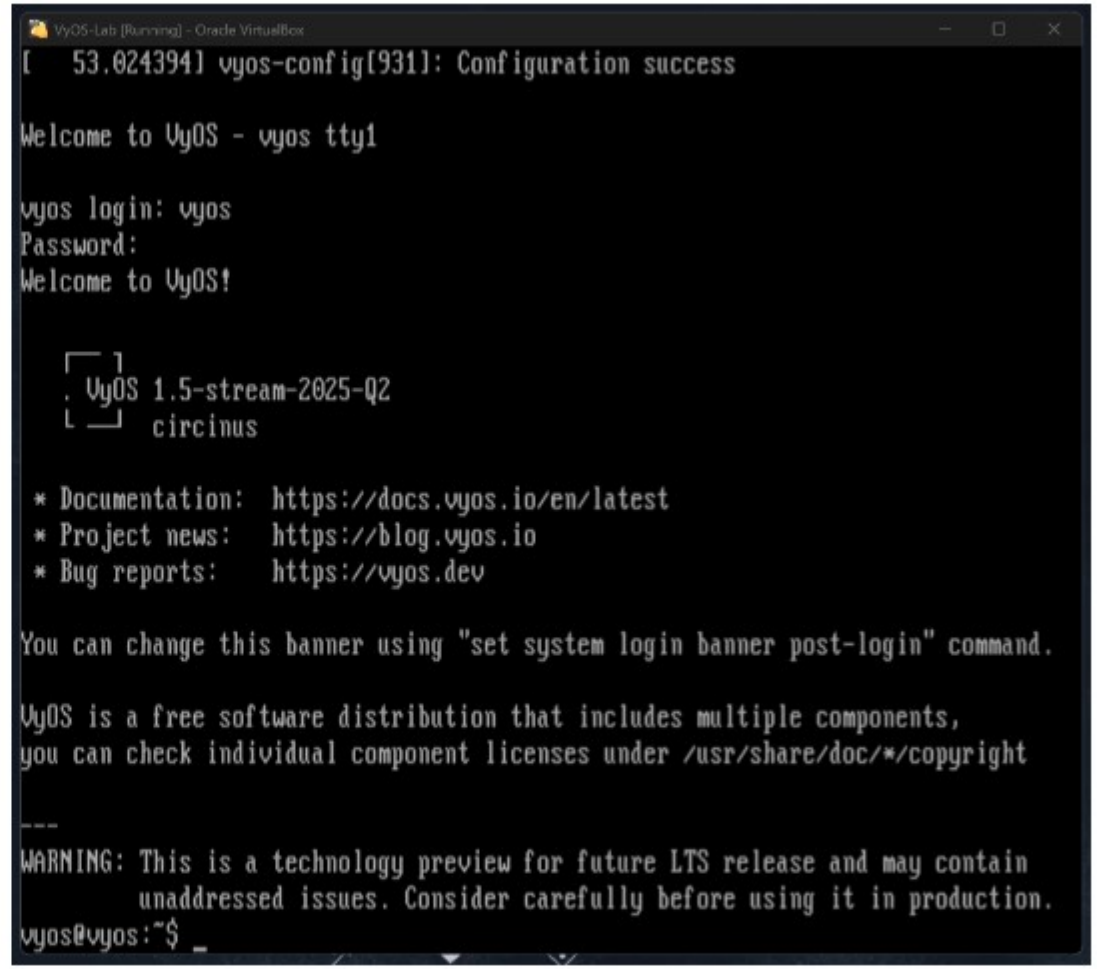
- Flood Web Server (port 80, hping3): successful



Smoothwall (free version) - Takeaway

- Strength:
  - Overall quite a good defence: matches desired network behaviour: blocks everything from WAN_NET.
- Limitation:
  - Once a rule explicitly permits traffic, firewall simply forwards it:
    - It's a border guard that does not patch vulnerabilities or secure internal services.
    - Implementing non-default features is essential.

VyOS


VyOS-Lab [Running] - Oracle VirtualBox
[53.024394] vyos-config[931]: Configuration success
Welcome to VyOS - vyos tty1
vyos login: vyos
Password:
Welcome to VyOS!
. VyOS 1.5-stream-2025-Q2
circinus
* Documentation: https://docs.vyos.io/en/latest
* Project news: https://blog.vyos.io
* Bug reports: https://vyos.dev
You can change this banner using "set system login banner post-login" command.
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
---
WARNING: This is a technology preview for future LTS release and may contain
unaddressed issues. Consider carefully before using it in production.
vyos@vyos:~$ _


VyOS-Lab [Running] - Oracle VirtualBox
vyos@vyos:~$ show firewall zone
Zone Members From Zone Firewall IPv4 Firewall IPv6
LAN eth0 WAN WAN-IN
LOCAL LOCAL LAN LAN-ROUTER
WAN LAN-ROUTER
WAN eth1 LAN LAN-OUT
vyos@vyos:~$



```
┌──(attackerpc㉿attackerpc)-[~]
└─$ ping -c 2 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.

--- 192.168.10.10 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1008ms
```

```
┌──(attackerpc㉿attackerpc)-[~]
└─$ hydra -l webserver -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.10 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations
, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-01 20:47:41
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.10.10:22/
[STATUS] 64.00 tries/min, 64 tries in 00:01h, 14344335 to do in 3735:31h, 4 active
[STATUS] 68.00 tries/min, 204 tries in 00:03h, 14344195 to do in 3515:45h, 4 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

```
attackerpc@attackerpc: ~
┌──(attackerpc㉿attackerpc)-[~]
└─$ nmap -sS 192.168.10.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-01 17:03 GMT
nass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns o
r specify valid servers with --dns-servers
Nmap scan report for 192.168.10.10
Host is up (0.0020s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 4.36 seconds

┌──(attackerpc㉿attackerpc)-[~]
└─$
```

```
35       accept    tcp               17     1020  tcp dport 22 ip saddr 192.168
.20.10  accept
```





```
Rulesets Information

-------------------------------
ipv4 Firewall "name LAN-OUT"

Rule     Action    Protocol      Packets    Bytes  Conditions
-------  --------  ----------  ---------  -------  ----------------------------
------------
10       accept    all                0        0  ct state { established, relat
ed }  accept
20       accept    all                0        0  accept
default  drop      all                0        0
```

```
vyos@vyos# set firewall
Possible completions:
 > bridge              Bridge firewall
+> flowtable           Flowtable
 > global-options      Global Options
 > group               Firewall group
 > ipv4                IPv4 firewall
 > ipv6                IPv6 firewall
+> zone                Zone-policy
```

```
vyos@vyos# set firewall ipvt name WAN-IN destination 192.168.10.10

  Configuration path: firewall [ipvt] is not valid
  Set failed

[edit]
vyos@vyos# _
```



OPNsense: Configuration & Setup

- FreeBSD-based open-source firewall with web GUI, PF packet filter, NAT, and integrated IDS/IPS (Suricata).
- Installed in VirtualBox; assigned interfaces; changed default credentials.
- Configured LAN/WAN rules to enforce desired network behaviour.
- Used NAT port forwarding to make LAN Web Server accessible from WAN for HTTP/SSH tests.
- Enabled Suricata intrusion detection on WAN with ET-Open rulesets.

Firewall Testing: Attack Scenarios

- Reconnaissance - ICMP ping from WAN blocked by default-deny rule.
- Port Scanning - Only HTTP (port 80) accessible; all other ports blocked.
- Vulnerability Scanning - Nmap revealed HTTP service; Suricata raised alerts for scans.
- Brute Force - SSH brute-force attempts from WAN blocked except Admin PC.
- DoS Attack - SYN flood caused GUI slowdown; port stayed open due to NAT forwarding.

OPNsense Overview

Strengths:

- Easy GUI setup and clear live logs.
- Effective filtering and NAT configuration.
- Integrated IDS provides attack visibility.

Limitations:

- IDS rules require careful tuning to detect all attacks.

Key Takeaway:

- OPNsense offers a strong open-source firewall with NAT and IDS/IPS



Conclusion

- Final Takeaway: All firewalls are secure when configured correctly, but their usability varies hugely.
- Best Firewall: OPNsense
  - Reason: Easiest to use with a modern Graphical User Interface (GUI) and includes powerful features like built-in intrusion detection.
- Worst Firewall (Most Challenging/Time Consuming): VyOS
  - Reason: Requires configuration using only a difficult and strict Command-Line Interface (CLI).
- Key Security Point: You must use extra features like rate-limiting to stop Brute Force and Denial-of-Service (DoS) attacks.